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PALAEOMAGNETISM OF MESOZOIC IGNEOUS ROCKS FROM THE MARANHÃO BASIN, BRAZIL, AND THE TIME OF OPENING OF THE SOUTH ATLANTIC

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From Middle-Upper Jurassic volcanics at the western margin of the Maranhão Basin (6.4°S, 47.4°W) 15 sites (121 samples) have a mean magnetization direction $D = 3.9°$, $I = -17.9°$ with $\alpha_{95} = 9.3°$, $k = 17.9$ after AF cleaning (all sites have normal polarity). This yields a pole (named SAJ_2) at 85.3°N, 82.5°E ($A_{95} = 6.9°$) which is near to the other known Middle Jurassic South American pole. For 21 sites (190 samples) from Lower Cretaceous basalt intrusions from the eastern part of the Maranhão Basin (6.5°S, 42°W) the mean direction is $D = 174.7°$, $I = +6.0°$ with $\alpha_{95} = 2.8°$, $k = 122$ (all sites have reversed polarity) yielding a pole (SAK_9) at 83.6°N, 261°E ($A_{95} = 1.9°$) in agreement with other Lower Cretaceous pole positions for South America. Comparing Mesozoic pole positions for South America and Africa in the pre-drift configuration after Bullard et al. [13] one finds a significant difference (with more than 95% probability) for the Lower Cretaceous and Middle Jurassic poles and also a probable difference for the mean Triassic poles indicating a small but probably stationary separation of the two continents from the pre-drift position in the Mesozoic until Lower Cretaceous time which may be due to an early rifting event.

1. Introduction

Most palaeomagnetic investigations on Jurassic and Cretaceous rocks in South America have been carried out in the southern part of the continent. In northern Brazil the intrusions and extrusions of the Maranhão Basin constitute igneous activity in different parts of the basin through most of the Mesozoic. From this area new Middle-Upper Jurassic and Lower Cretaceous palaeopoles are presented in this paper. Mesozoic palaeomagnetic data are of importance for our knowledge of the initial opening of the South Atlantic.

2. Geological environment and sampling sites

At the western border of the Maranhão sedimentary basin the basaltic rocks occur as extensive flows and sills while intrusive forms predominate at its eastern border (Fig. 1). According to Sial et al. [1] the Maranhão basalts are essentially oversaturated tholeiites.

In the areas which have been sampled (Fig. 1) there exist some K-Ar ages. Four samples from near Porto Franco give an average age of 158 ± 12 m.y. (standard deviation) [1]. Recent determinations by Hasui et al. [2] also yield an average age of 160 m.y. which is Middle-Upper Jurassic. In the second area (Teresina, Floriano and Picos, see Fig. 1) the mean age for 8 samples is 118 ± 6 m.y. or Lower Cretaceous [3] [4].

The samples were cored (diameter 2.5 cm) with a portable drilling machine along roads, rivers and in quarries. In the surroundings of Porto Franco 16 sites (flows and intrusions) in an area of about 60 km in diameter were sampled (Fig. 1 and Table 1). In the

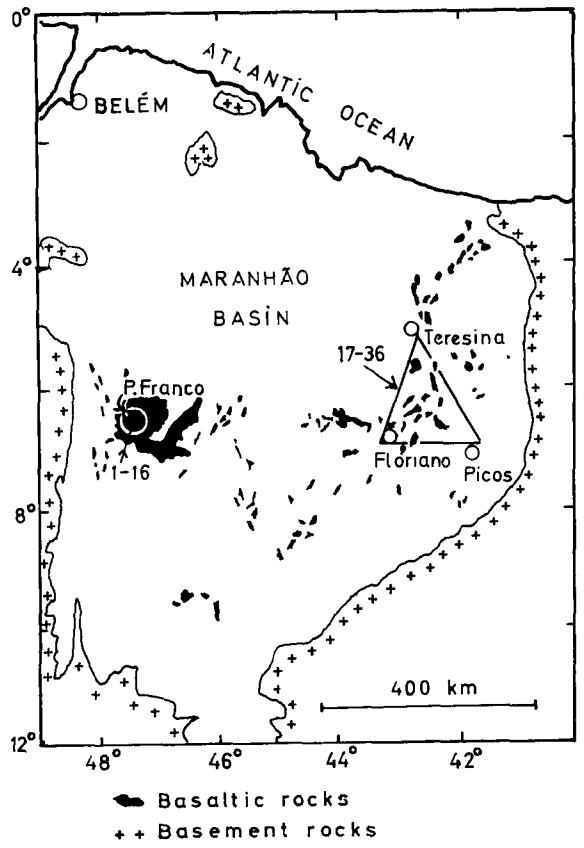


Fig. 1. Map showing sampling sites in the Maranhão Basin (after Sial et al. [1]).

other area between Teresina, Floriano and Picos 22 sites (probably all intrusions) were sampled (Table 2). Both areas are tectonically undisturbed, therefore no bedding correction was necessary.

3. Measurements and results

The natural remanent magnetization (NRM) of all samples was measured with a simple astatic magnetometer or with a "Digico" spinner magnetometer. Pilot samples of all sites were then demagnetized in steps up to 1000 Oe AF peak values and, according to these results, all samples were treated with the appropriate alternating field (Tables 1 and 2) and the characteristic remanence (CARM) was measured with the spinner magnetometer.

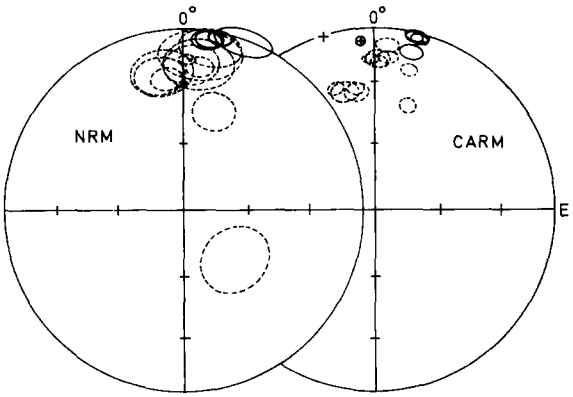


Fig. 2. Equal area projection of site means for Middle-Upper Jurassic rocks from Porto Franco (only the respective circles of confidence are shown; broken-line curves for negative inclination). Cross and cross-with-circle (positive inclination) denote present earth's field direction and central dipol field direction respectively.

3.1. Middle to Upper Jurassic basaltic rocks from Porto Franco

The results are summarized in Table 1 and the circles of confidence of the site-mean directions of NRM and CARM are shown in Fig. 2. The within-site scatter as well as the between-site scatter could be reduced by the AF treatment. For one site no consistent result could be obtained. The intensity of the NRM varied between 0.5 and 5×10^{-3} G (10^{-3} G $\hat{=}$ 1 A m^{-1}) and was reduced to about 30% by the applied alternating field. All sites have normal direction of magnetization. Giving unit weight to each site the mean direction ($D = 3.9°$, $I = -17.4°$ with $\alpha_{95} = 9.3$, $k = 17.9$, $N = 15$) is significantly different from the present local field or dipole field at the sampling site (Fig. 2). The angular standard variation ($s \approx 81 \cdot k^{-0.5}$) equals 19.1° and indicates the usual palaeo secular variation for that latitude [5]. However, the distribution shows some degree of ovalness as can be seen in Fig. 2 or in Fig. 5 where the virtual geomagnetic poles (VGP) for the individual sites are plotted. The VGP's are roughly distributed along the 0°/180° meridian. It is interesting that the other Middle Jurassic palaeomagnetic result for South America [6] has similar features.

TABLE 1

Palaeomagnetic results for Middle-Upper Jurassic volcanic rocks from Porto Franco. Mean site location is 6.4°S, 47.4°W

Site	NRM					AF(Oe)	CARM					VGP	
	N	D (°)	I (°)	α_{95} (°)	k		N	D (°)	I (°)	α_{95} (°)	k	°N	°E
1	9	1.9	−18.3	2.8	337	240	9	358.9	−17.8	2.5	425	86.9	152
2	5	6.3	−14.3	15.5	26	200	5	4.7	−9.9	4.1	344	85.0	26
3	12	8.2	+6.4	5.2	70	300	12	13.7	+11.4	4.5	91	71.6	1
4	7	6.7	−21.7	5.0	149	400	7	2.1	−19.5	2.9	425	85.7	102
5	9	10.0	−10.8	11.0	23	260	8	5.2	−16.7	3.9	193	84.4	65
6	7	no consistent result											
7	7	6.4	−19.8	13.2	22	330	8	14.1	−21.7	3.4	256	75.2	62
8	10	19.9	+0.4	10.1	24	300	10	14.3	+1.5	4.0	144	73.9	16
9	7	349.8	−26.1	10.5	34	330	12	342.0	−30.6	3.4	162	69.8	191
10	5	17.9	−42.5	9.0	73	180	9	17.8	−39.4	4.0	164	66.7	87
11	6	133.7	−56.3	15.9	19	700	8	350.9	−34.5	5.1	121	74.5	167
12	5	352.6	−28.2	6.0	161	350	11	351.5	−32.7	3.7	150	76.0	168
13	6	351.2	−24.5	12.4	30	300	6	345.0	−34.1	5.9	129	70.9	181
14	6	3.3	−19.4	11.0	38	400	5	1.2	−17.9	4.7	261	86.3	110
15	7	12.8	+3.6	3.6	272	300	7	15.5	+3.4	3.3	324	72.4	16
16	5	8.7	+5.8	4.8	254	250	4	14.3	+2.6	2.5	1310	73.8	16
Mean of site means:							15	3.9	−17.9	9.3	17.9	85.3	82.5
												A_{95} = 6.9°	K = 31.8

N = number of samples, D = declination, I = inclination, α_{95} or A_{95} = radius of circle of confidence at the 5% significance level, k or K = precision parameter, AF = peak value of alternating field cleaning (10^4 Oe ≙ 1 T).

TABLE 2

Palaeomagnetic results for Lower Cretaceous volcanic rocks from Teresina, Floriano and Picos. Mean site location is 6.5°S, 42°W

Site	NRM					AF(Oe)	CARM					VGP	
	N	D (°)	I (°)	α_{95} (°)	k		N	D (°)	I (°)	α_{95} (°)	k	°N	°E
17	10	180	+17.7	5.5	76	250	10	178.0	+14.9	2.5	365	87.5	193
18	9	181	+16.0	3.0	290	250	9	180.2	+13.2	1.4	1220	89.4	116
19	8	171	+19.4	4.0	192	300	8	175.2	+15.3	3.1	310	84.9	209
20	10	252	+42.8	31.5	3.3	330	11	179.7	+17.0	6.2	54	87.4	145
21	7	100	−26.1	8.8	47		no consistent result						
22	6	167	+13.7	4.7	202	200	6	173.9	+16.7	4.0	268	83.5	206
23	9	176	−16.9	5.9	77	450	8	177.1	−2.3	1.1	2310	82.5	294
24	7	173	−6.0	2.9	421	200	7	174.1	−6.2	1.7	1180	79.4	283
25	6	175	−5.4	18.4	14	300	5	175.8	+4.0	1.7	1860	84.1	271
25B	8	185	−12.4	16.1	13	300	8	174.3	+7.3	3.0	337	83.8	250
26	4	177	+6.1	4.3	447	300	16	166.0	+6.4	3.5	109	75.6	241
26B	4	182	+6.3	5.4	267	300	8	181.6	+8.8	2.9	358	87.1	351
27	8	166	−18.5	8.8	40	270	8	172.4	−1.3	2.6	434	79.3	271
28	7	195	+1.7	4.4	186	220	7	176.4	+4.4	3.0	393	84.1	279
29	6	173	−2.9	3.6	341	240	6	174.4	−0.2	2.1	939	81.1	278
30	8	172	+7.1	2.1	682	320	8	172.3	+8.2	2.6	451	81.8	246
31	9	172	−1.3	5.1	102	350	8	172.9	+3.5	1.6	1065	81.3	262
32	10	169	−2.1	5.2	87	200	10	175.1	+3.2	1.7	787	82.7	276
33	5	172	+6.0	3.4	500	300	13	170.5	+6.1	1.8	526	79.7	250
34	8	178	+1.8	2.4	524	250	10	174.7	+3.8	2.2	463	82.6	272
35	4	180	−3.4	24.0	15	300	16	173.5	+5.2	4.1	82	82.1	262
36	4	166	−13.9	37.0	7	200	8	172.4	−0.1	5.5	99	79.6	271
Mean of site means:							21	174.7	+6.0	2.8	122	83.6	261
												A_{95} = 1.9°	K = 276

Legend see Table 1.

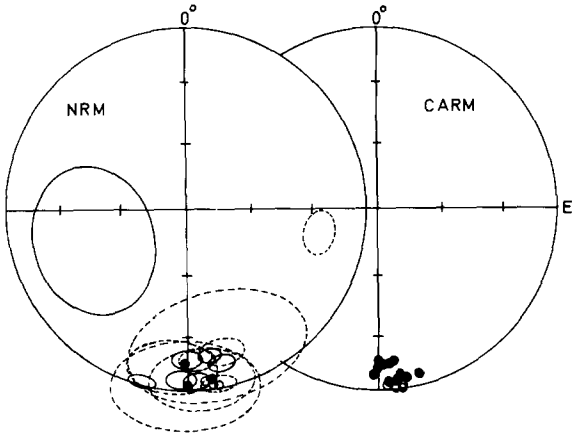


Fig. 3. Site means or the respective circles of confidence for Lower Cretaceous rocks from Teresina, Floriano and Picos. Broken-line curves and open circles for negative inclination.

3.2. Lower Cretaceous basaltic rocks from Teresina, Floriano and Picos

The palaeomagnetic results are listed in Table 2 and shown in Fig. 3. The within-site scatter of the NRM which is large in some cases could be drastically reduced by AF cleaning. Typical examples are shown in Fig. 4. The intensity of the NRM ($1-10 \times 10^{-3}$ G) was reduced to 50–10% by the applied alternating field. The remanence of one site was extremely unstable giving no consistent result after AF cleaning. Giving unit weight to each site the mean direction is $D = 174.7°$, $I = +6.0°$ with $\alpha_{95} = 2.8°$, $k = 122$, $N = 21$. All sites have reversed directions. The angular standard deviation ($s = 7.3°$) is relatively small but it can be concluded with some confidence that several sites sampled belong to the same intrusion. The sites are situated in an area of about 200 km in diameter and those sites which are closer together have in most cases significantly different magnetization directions.

The VGP's for the sites, the mean poles and the respective circles of confidence are shown in Fig. 5.

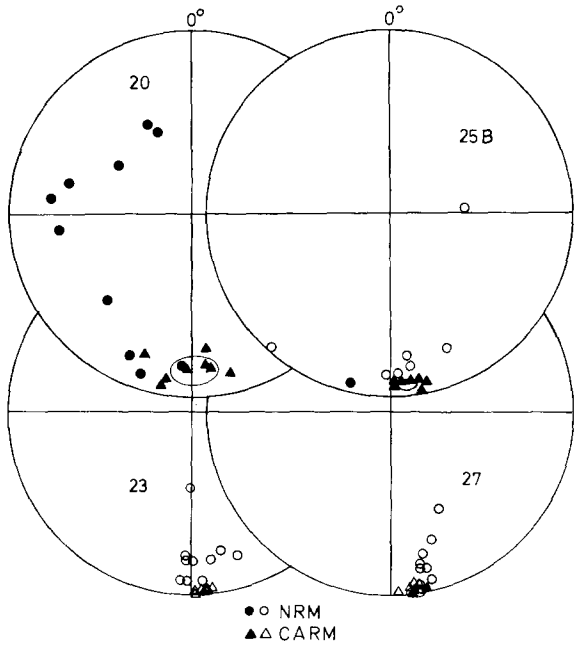


Fig. 4. Typical AF demagnetization result for samples. Open symbols mark negative inclination, oval curves denote the circles of confidence of mean CARM directions. Numbers refer to site numbers.

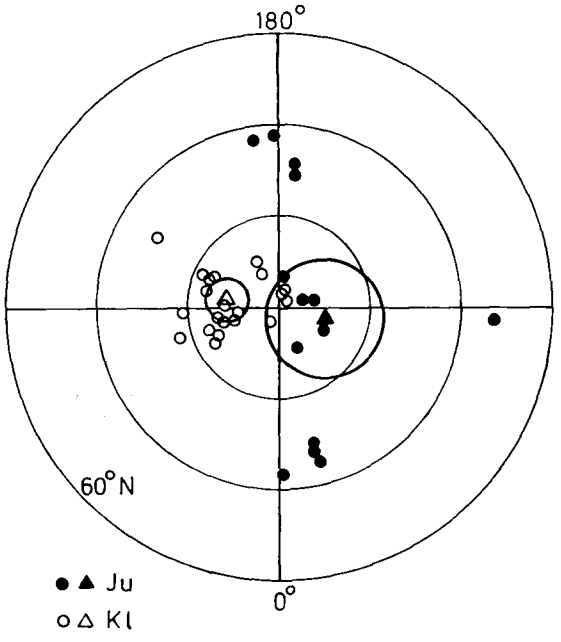


Fig. 5. Virtual geomagnetic poles for the sites (circles) and mean palaeomagnetic poles (triangles) with circles of confidence.

Both pole positions (85°N, 83°E with A_{95} = 6.9° for the Middle to Upper Jurassic rocks and 84°N, 261°E with A_{95} = 1.9 for the Lower Cretaceous rocks) are significantly different from each other and also different from the present geographic pole. Following the nomenclature which has been used by several authors the poles have been named SAJ_2 and SAK_9 respectively. The polarities of the two collections are consistent with the polarity time scale of Larson et al. [7]: There is a relatively long period of reversed polarity around 115 m.y. B.P. (to compare with an age of about 118 m.y. for the reversed sequence) and a long period of normal polarity before 153 m.y. B.P. (to compare with an age of about 158 m.y. for the normal sequence).

4. Synthesis of palaeomagnetic data and the time of opening of the South Atlantic

In Table 3 and Fig. 6 South American Mesozoic poles are compiled. The new pole SAJ_2 is in agreement with the Middle Jurassic pole SAJ_1 from Vilas [6]. The other pole SAK_9 is very near to the pole

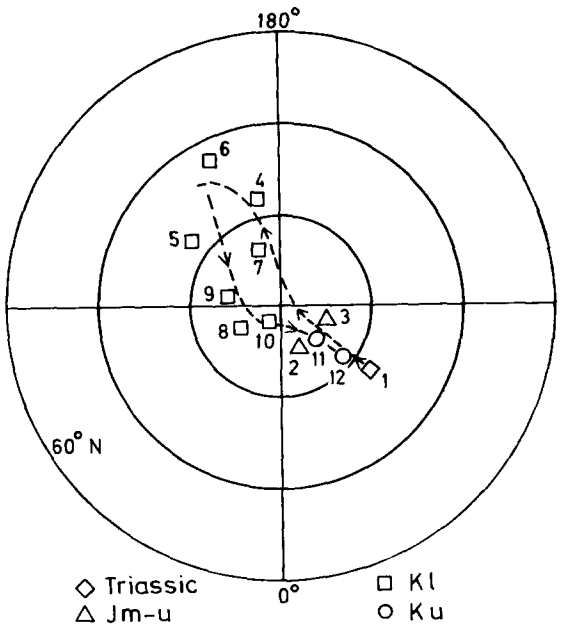


Fig. 6. Mesozoic palaeomagnetic poles for South America in its present-day position and proposed polar wander path. Labels refer to Table 3.

TABLE 3

Mesozoic palaeomagnetic poles for South America

Label	Formation	Name	Age (m.y.)	Pole		A_{95} (°)	Rotated poles				Reference
				°N	°E		°N,	°E [1]	°N,	°E [2]	
1	South America	Triassic mean		78	56	6.3	61	253	70	247	*
2	Chon-Aike	SAJ_1	157–173 Jm	85	17	6	55	263	63	261	[6]
3	Maranhão	SAJ_2	158 ± 12 Jm-u	85	83	6.9	53	255	62	250	this paper
4	Rio de los Molinos	SAK_3	129–150 Kl	78	193	8	39	253	48	249	[21]
5	Serra Geral	SAK_1	115–130 Kl	78	234	5.7	38	264	47	261	[22]
6	Almafuerte	SAK_5	123 ± 4 Kl	72	205	6	32	255	41	251	[23]
7	C. Colorado	SAK_2	121 ± 3 Kl	83	196	10	43	256	53	252	[23]
8	Serra Geral	SAK_8	119 ± 5 Kl	85	295	3.7	48	268	57	266	[8]
9	Maranhão	SAK_9	118 ± 6 Kl	84	261	1.9	45	266	54	264	this paper
10	C. Rumipalla	SAK_4	<121 Kl	88	326	9	50	263	60	261	[23]
11	Pirgua	SAK_6	77–114 Ku	85	42	8	55	260	82	241 [3]	[23]
12	Poços de Caldas	SAK_7	63–80 Ku	81	53	10	59	256	85	211 [3]	[24]

Rotation parameters: [1] From present to pre-drift position: 57° about a pole at 44°N, 30.6°W [13]. [2] From present to the position estimated for 110 m.y. B.P.: 48.3° about a pole at 49.2°N, 31.8°W [15]. [3] From present to the position estimated for 80 m.y. B.P.: 32.8° about a pole at 67.3°N, 39.5°W [19].

* Mean of 12 Permo-Triassic and Triassic poles compiled by Hargraves [20].

SAK_8 which was recently determined by Pacca et al. [8] for Serra Geral basalts of the same age. In Fig. 6 also an apparent polar wander path for South America in its present position is proposed. Starting with the Triassic there seems to be an excursion of about 25° between Upper Jurassic and Lower Cretaceous with a return shift until Upper Cretaceous time. Considering the uncertainty of the pole positions and the age determinations details of this apparent polar wander path cannot be resolved.

The time of opening of the South Atlantic has been discussed by numerous authors. In most of the recent publications an age of 140 to 130 m.y. (between Upper Jurassic and Lower Cretaceous) is assumed for the initial rifting based on geological [9] or palaeobiographic [10] evidence or derived from the interpretation of magnetic lineations in the southern South Atlantic [11]. An early beginning of rifting in Jurassic time or earlier has also been discussed [12]. However, reconstructions with a significant separation of South America from Africa from the pre-drift configuration after Bullard et al. [13] have not been suggested for ages older than 112 m.y. B.P. [11,14] or 110 m.y. B.P. [15]. Against this background the "rotated" pole positions for South America for the pre-drift configuration [13] and for the 110-m.y. reconstruction [15] given in Table 3 can be compared with African poles (Table 4, Africa in its present position). It is illustrated in Fig. 7 that the Lower Cretaceous pole positions are brought into closer proximity by using the reconstruction proposed for 110 m.y. In order to test the significance an F-test on mean values [16] was used: Mean Lower Cretaceous poles for South America in the Bullard pre-drift configuration and for Africa were calculated giving unit weight to the respective poles (poles 4–10 from Table 3 and 6–9 from Table 4). This yields a mean South American pole at 260.4°E, 42.2°N ($N = 7, R = 6.9466, K = 112, A_{95} = 5.7°$) and an African pole at 259.6°E, 53.6°N ($N = 4, R = 3.9559, K = 68, A_{95} = 11.2°$). The value of the test variable F (= 4.61) is greater than the critical value at the 5% significance level ($F_{0.05} = 3.5$). Therefore the probability is less than 5% that the pole positions are the same and it can be concluded that the Lower Cretaceous poles of both continents in the pre-drift configuration differ significantly whereas they do not differ for the 110-m.y. B.P. reconstruction (the last result is clear from Fig. 7 and needs no further statistical test). This

TABLE 4

Mesozoic palaeomagnetic poles for Africa

Label	Formation	Age (m.y.)	Pole		A_{95} (°)	Reference
			°N	°E		
1	Africa	Triassic mean	69	263	4.9	*
2	Karroo	154–190	69	263	8	[25]
3	Stormberg	J	71	269	15	[25]
4	Hoachanas	161–173	62	252	7	[17]
5	Mateke Hills Complex	~168	59	260	8.3	[26]
6	Moroccan volcanics	Kl	44	251	10	[27]
7	Mlanje Massif	116–128	60	262	12	[28]
8	Kaoko lavas	110–128	48	267	2	[17]
9	Lupata volcanics	106–111	62	260	3.5	[29]
10	Moroccan sediments	Kl-u	75	217	5.5	[30]
11	Nubian sandstone	Ku	79	220	4.0	[31]
12	Wadi Natash volcanics	81–90	64	218	4.2	[32]
13	Kimberlite pipes	83–89	58	237	15.3	[33]
14	Red siltstone	Ku	79	208	6	[29]
15	Volcanics Sicily	71–81	63	229	3.8	[34]
16	Tororo	Ku(?)	76	195	9	[35]

* Mean of 7 Triassic palaeomagnetic poles from a compilation by McElhinny and Brock [36].

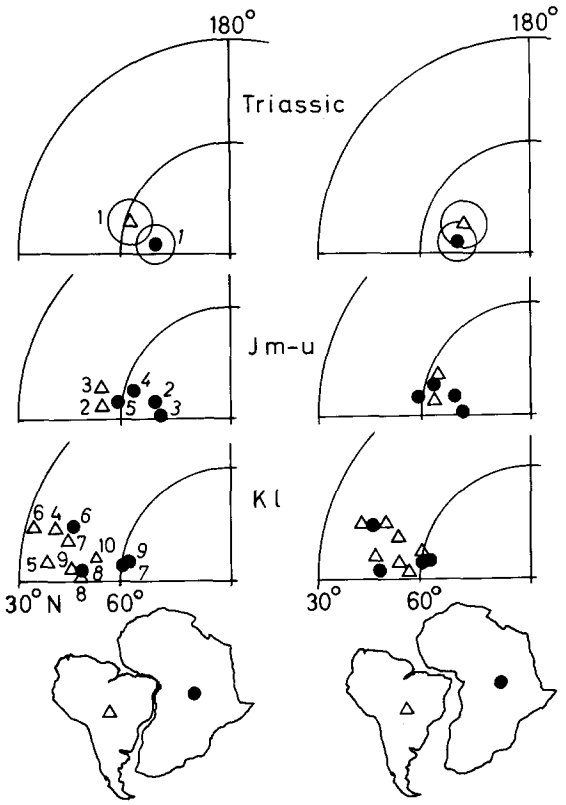


Fig. 7. Comparison between South American and African Mesozoic poles in the pre-drift configuration [13] (left) and for a reconstruction estimated for 110 m.y. B.P. [15] (right). Africa in its present position. Labels refer to poles in Tables 3 and 4 (African poles in italics). For rotation parameters see Table 3.

finding implies that the separation of the continents had already begun at the time of emplacement of the Lower Cretaceous formations which is, however, earlier than the suggested date of the reconstruction. A similar result was obtained by Gidskehaug et al. [17] using the reconstruction from Le Pichon and Hayes [14] for an assumed age of 112 m.y. [11]. (Rotation of 4.6° of South America about 21.5°N, 14°W from the pre-drift configuration. The respective South American pole would be situated about 4° more to the south in Fig. 7.)

Going back in time to the Middle to Upper Jurassic and Triassic time there is again the same tendency as for Lower Cretaceous time that in the pre-drift configuration the South American poles are situated more to the south (relative to the grids in Fig. 7) than the corresponding African poles. Using the same reconstruction for 110 m.y. B.P. as before the agreement between the pole positions becomes better. For a significance test again mean Middle Jurassic poles were calculated for South America and Africa in the pre-drift configuration yielding poles at 258.9°E, 54°N ($N = 2, R = 1.9980, K = 504, A_{95} = 11.1$; poles 2 and 3 from Table 3) and at 260.1°E, 65.3°N ($N = 4$, $R = 3.9815, K = 162, A_{95} = 7.2$; poles 2–5 from Table 4). The value of F (= 5.05) is greater than the critical one ($F_{0.05} = 4.6$) indicating that also the Middle Jurassic poles of both continents in the pre-drift configuration differ (i.e. different with more than 95% probability, but this seems not so well demonstrated as for the Lower Cretaceous because of less data). For the Triassic the pre-drift mean poles (see Tables 3 and 4) are not different on the 5% significance level. F (= 3.0) is smaller than the critical value ($F_{0.05} = 3.25$). On the other hand we have the feeling that these poles may differ significantly if more data would be available because F is near the critical value and the sense of deviation of the mean pole positions is similar as for the Middle Jurassic and Lower Cretaceous poles. Consequently for the Middle Jurassic and – with restrictions concerning the significance – also for the Triassic the palaeomagnetic data indicate that South America and Africa were not in the pre-drift configuration but the respective coastlines in the South Atlantic were separated a few hundred kilometers more than in the reconstruction after Bullard et al. [13] in a sense as in the assumed reconstruction for 110 m.y. B.P. (not necessarily this particular reconstruction because it is not uniquely determined by the pole positions). In Fig. 7 it is also demonstrated that no change in the separation of the two continents from Triassic to Lower Cretaceous time (i.e. a progressing sea-floor spreading) can be derived from the data. The conclusion is therefore: The palaeomagnetic data indicate a small separation of South America and Africa from the so-called pre-drift configuration in Lower Cretaceous, Middle Jurassic and probably also in Triassic time. Assuming the validity of the used pre-drift reconstruction used [13] this separation may be due to an early rifting in pre-Triassic or at least in Early Jurassic time which was then more or less stationary during most of the Mesozoic until the end of Lower Cretaceous time. In some early palaeomagnetic

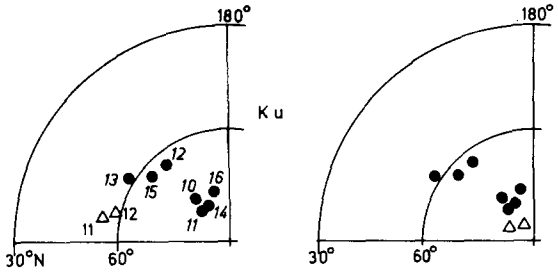


Fig. 8. Comparison of South American (triangles) and African (circles) Upper Cretaceous poles in the pre-drift configuration [13] (left) and for a reconstruction estimated for 80 m.y. B.P. [19] (right). Labels refer to Tables 3 and 4.

investigations a similar timing of the beginning of rifting has been suggested (e.g. [18]).

Poles for the Upper Cretaceous are plotted in Fig. 8 in pre-drift configuration and for a reconstruction for 80 m.y. B.P. [19]. For the pre-drift configuration (and also for the 110-m.y. B.P. reconstruction) South American and African pole positions differ, consistent with a quite significant and fast opening of the South Atlantic during Upper Cretaceous time [14,15,19]. The reconstruction for 80 m.y. B.P. brings the South American Upper Cretaceous poles near to the geographic pole (see Fig. 8). Some recent determinations of Upper Cretaceous African poles (Table 4) yield also positions near the geographic pole (and near the rotated South American poles) whereas other poles are situated at lower latitudes. Probably the time density of Upper Cretaceous palaeomagnetic data is at present not good enough to define a polar wander path for a time of fast opening of the South Atlantic.

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References

1 A.N. Sial, D.R. Pessoa and K. Kawashita, New potassium-argon ages, strontium isotope ratio measurement and chemistry of Mesozoic basalts, state of Maranhão, northeast Brazil, Bol. Mineral., Recife 4 (1976) 59–72.
2 Y. Hasui, W.T. Hennies and W. Iwanuch, Idade do magmatismo básico mesozoico do Brasil Central, Resumo do Trabalhos, 29th Congr. Bras. Geol., Belo Horizonte (1976) 181–182.
3 U.G. Cordani, Idade do vulcanismo no Oceano Atlantico Sul, Bol. IGA 1 (1970) 9–75.
4 A.B. Nunes, R.F. Lima and B.F. Negreiros, Geologia da folha SB 23 Teresina e parte da folha SB 24 Jaguaribe, Proj. RADAM 2 (1973) 1–32.
5 A. Brock, An experimental study of palaeosecular variation, Geophys. J. R. Astron. Soc. 24 (1971) 303–317.
6 J.F. Vilas, Palaeomagnetism of some igneous rocks of the Middle Jurassic Chon-Aike formation from Estancia La Reconquista, Province of Santa Cruz, Argentina, Geophys. J. R. Astron. Soc. 39 (1974) 511–522.
7 R.L. Larson and T.W.C. Hilde, A revised time scale of magnetic reversals for the Early Cretaceous and Late Jurassic, J. Geophys. Res. 80 (1975) 2586–2594.
8 I.G. Pacca and F.Y. Hiodo, Palaeomagnetic analysis of Mesozoic Serra Geral basaltic lava flows in southern Brazil, An. Acad. Bras. Cienc. (Supl.) 48 (1975) 207–214.
9 T.H. Van Andel, J. Thiede, J.G. Sclater and W.W. Hay, Depositional history of the South Atlantic Ocean during the last 125 million years, J. Geol. 85 (1977) 651–698.
10 R. Förster, Evidence for an open seaway between northern and southern proto-Atlantic in Albian times, Nature 272 (1978) 158–159.
11 R.L. Larson and J.W. Ladd, Evidence for the opening of the South Atlantic in the Early Cretaceous, Nature 246 (1973) 209–212.
12 X. Le Pichon, Sea-floor spreading and continental drift, J. Geophys. Res. 73 (1968) 3661–3697.
13 E.C. Bullard, J.E. Everett and A.G. Smith, The fit of the continents around the Atlantic, Philos. Trans. R. Soc. London, Ser. A, 258 (1965) 41–75.
14 X. Le Pichon and D.E. Hayes, Marginal offsets, fracture zones and the early opening of the South Atlantic, J. Geophys. Res. 76 (1971) 6283–6293.
15 J.G. Sclater, S. Hellinger and C. Tapscott, The paleobathymetry of the Atlantic Ocean from the Jurassic to the present, J. Geol. 85 (1977) 509–552.
16 G.S. Watson, Analysis of dispersion on a sphere, Mon. Not. R. Astron. Soc., Geophys. Suppl. 7 (1956) 153–159.
17 A. Gidskehaug, K.M. Creer and J.G. Mitchell, Palaeomagnetism and K-Ar ages of the South-West African basalts and their bearing on the time of initial rifting of

the South Atlantic Ocean, Geophys. J. R. Astron. Soc. 42 (1975) 1–20.
18 K.M. Creer, Paleomagnetic data from the Gondwanic continents, Philos. Trans. R. Soc. London, Ser. A, 258 (1965) 27–40.
19 J. Francheteau, Plate tectonics model of the opening of the Atlantic Ocean south of the Azores, in: Implications of Continental Drift to the Earth Sciences, Vol. 1, D.H. Tarling and S.K. Runcorn, eds. (Academic Press, London, 1973) 197–202.
20 R.B. Hargraves, Problems in palaeomagnetic synthesis illustrated by results from Permo-Triassic dolerites in Guyana, Phys. Earth Planet. Inter. 16 (1978) 277–284.
21 E. Linares and D.A. Valencio, Palaeomagnetism and K-Ar ages of some trachybasaltic dikes from Rio de los Molinos, Province of Córdoba, Argentina, J. Geophys. Res. 80 (1975) 3315–3321.
22 K.M. Creer, Palaeomagnetism of the Serra Geral Formation, Geophys. J. R. Astron. Soc. 7 (1962) 1–22.
23 D.A. Valencio and F.A. Vilas, Sequence of the continental movements occurred prior to and after the formation of the South Atlantic, An. Acad. Bras. Ciênc. (Supl.) 48 (1976) 377–386.
24 N.D. Opdyke and W.D. MacDonald, Palaeomagnetism of Late Cretaceous Poços de Caldas Alkaline Complex, southern Brazil, Earth Planet. Sci. Lett. 18 (1973) 37–44.
25 A. Hicken, E. Irving, L.W. Law and J. Hastie, Catalogue of Palaeomagnetic Directions and Poles (first issue), Department of Energy, Mines and Resources, Ottawa, Publ. Earth Phys. Branch, Vol. 45, No. 1 (1972).
26 D.I. Gough, A. Brock, D.L. Jones and N.D. Opdyke, The paleomagnetism of the Ring Complexes at Marangudzi and the Mateke Hills, J. Geophys. Res. 69 (1964) 2499–2507.
27 C. Bardon, A. Bossert, R. Hamzeh, J.P. Rolley and M. Westphal, Etude paléomagnetique de formations volcaniques du Cretace inferieur dans l'Atlas de Beni Mellal (Maroc), C.R. Acad. Sci., Paris, Ser. D, 227 (1973) 2141–2444.
28 J.C. Briden, A new palaeomagnetic result from Lower Cretaceous of East-Central Africa, Geophys. J. R. Astron. Soc. 12 (1967) 375–380.
29 D.I. Gough and N.O. Opdyke, The palaeomagnetism of the Lupata Alkaline Volcanics, Geophys. J. R. Astron. Soc. 7 (1963) 457–468.
30 E.A. Hailwood, The palaeomagnetism of Triassic and Cretaceous rocks from Marocco. Geophys. J.R. Astron. Soc. 41 (1975) 219–235.
31 A. Schult, H.C. Soffel and A.G. Hussain, Palaeomagnetism of Cretaceous Nubian Sandstone, Egypt, J. Geophys. 44 (1978) 333–340.
32 E.M. El Shazly and M. Krs, Palaeography and palaeomagnetism of Nubian Sandstone, Eastern Desert of Egypt, Geol. Rundsch. 62 (1973) 212–225.
33 P.L. McFadden and D.L. Jones, The palaeomagnetism of some Upper Cretaceous Kimberlite Occurences in South Africa, Earth Planet. Sci. Lett. 34 (1977) 125–135.
34 A. Schult, Palaeomagnetism of Upper Cretaceous volcanic rocks in Sicily, Earth Planet. Sci. Lett. 19 (1973) 97–100.
35 P.K.S. Raja and J.B. Vise, Palaeomagnetism of the Tororo Ring Complex, S.E. Uganda, Earth Planet. Sci. Lett. 19 (1973) 438–442.
36 N.W. McElhinny and A. Brock, A new palaeomagnetic result from East Africa and estimates of the Mesozoic palaeoradius, Earth Planet. Sci. Lett. 27 (1975) 321–328.